

Ravenswood School 4^{unit} 1999 3 hours.

Question 1

- a) Reduce the complex expression $\frac{(2-i)(8+3i)}{(3+i)}$ to the form $a+ib$ where a and b are real numbers.
- b) The complex number z is given by $z = -\sqrt{3} + i$
- Write down the values of $\arg z$ and $|z|$
 - Hence or otherwise show that $z^7 + 64z = 0$
- c) Find the roots of the equation $(2+i)z^2 - 4z + (2-i) = 0$ expressing any complex roots in the form $a+ib$ where a and b are real.

Question 2 (begin a new page)

- a) Given that $P(x) = (x^4 - 1)(x^2 - 2)$ factorise $P(x)$ completely over:
- the rational numbers.
 - the real numbers.
 - the complex numbers.
- b) If the polynomial $P(x) = x^4 + x^2 + 6x + 4$ has a rational zero of multiplicity 2, find all the zeros of $P(x)$ over the complex field.
- c) Consider the polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$
- If $P(x)$ has roots $(a+bi)$, $(a-2bi)$ where a and b are real find the values of a and b .
 - Hence find the zeros of $P(x)$ over the complex field and express $P(x)$ as the product of two quadratic factors.

Question 3 (begin a new page)

Pi

- a) Let α, β, γ be the roots of the polynomial $x^3 + 4x^2 - 3x + 1 = 0$

Find the equations with roots:

- $2\alpha, 2\beta, 2\gamma$
 - $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$
- b) Graph the function $f(x) = 1 - x^2$ for $-2 \leq x \leq 2$
- Without using calculus, neatly sketch the following curves, clearly showing their main features. Use half a page for each graph.
- $y = |f(x)|$
 - $|y| = f(x)$
 - $y = \{f(x)\}^2$
 - $y = e^{f(x)}$

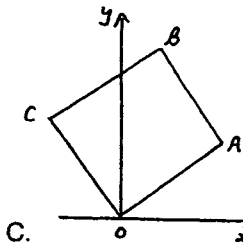
Question 4 (begin a new page)

- a) Given that $1, w$ and w^2 are the cube roots of unity, the roots of $z^3 = 1$ simplify $(1-w)(1-w^2)((1-w^4)(1-w^8))$

- b) Sketch the following loci on separate Argand diagrams:

i) $\arg(z+1+i) = \frac{\pi}{4}$ ii) $|z-2| = |z+ii|$

- c) OABC is a square in the complex plane and the point A represents the complex number z .



- State the complex numbers represented by B and C.
- Draw the square reflected in the x axis to become $OA'B'C'$
What complex numbers are represented by A', B', C' .

P

- $x^2 - 3x + k - 2 = 0$ has two distinct real roots.

- How much money would now be in the fund at the end of 20 years?

-
- A square \$ABCD\$ is shown with vertices \$A\$ (top-left), \$B\$ (bottom-left), \$C\$ (bottom-right), and \$D\$ (top-right). Point \$E\$ is on side \$AD\$ and point \$F\$ is on side \$DC\$. A line segment \$EF\$ is drawn such that \$EF \parallel AC\$. The triangle \$EBF\$ is shaded with horizontal lines. The area of the shaded triangle is \$\frac{1}{8}\$ of the area of the square.

- quadrilateral EFCD is given by $A = \frac{1}{2} (4 + 2x - x^2)$

- $$f(x) > 0, \quad f'(x) < 0, \quad f''(x) > 0$$

- ii) State the least value of $f(x)$ in this interval.

P

- c) Find the volume obtained when the area in the first quadrant enclosed by the curves $y = \sin x$ and $y = \cos x$ and the y axis is rotated about the x axis.

iv) at least 1 failure.

- | | | | | | |
|------|---|------|------|------|------|
| x | 0 | 0.2 | 0.4 | 0.6 | 0.8 |
| f(x) | 0 | 0.20 | 0.39 | 0.56 | 0.71 |

Use Simpson's Rule with 5 function values to calculate $\int_0^{0.8} f(x) dx$

Question 8 (begin a new page)

a) Draw neat sketch graphs of the following showing their main features

i) $y = 2 - \sin x$

ii) $y = \ln(2x - 6)$

b) Express $\frac{x^2 - 4x - 1}{(1+x^2)(1+2x)}$ as the sum of two partial fractions.

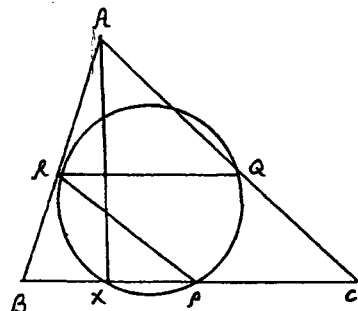
Hence find $\int \frac{x^2 - 4x - 1}{(1+x^2)(1+2x)} dx$

c) P, Q, R are the mid-points of the sides BC, CA and AB of a triangle ABC. The circle through P, Q and R meets the three sides again at X, Y and Z respectively. show that:

i) RPCQ is a parallelogram.

ii) Triangle XQC is isosceles.

iii) AX is perpendicular to BC



END OF PAPER

1999 4 JUNE 2 Yearly Solutions

Q.1. a) $\frac{(2-i)(8+3i)}{3+i} = \frac{16+6i-8i-3i^2}{3+i}$
 $= \frac{19-2i}{3+i} \times \frac{3-i}{3-i}$
 $= \frac{57-19i-6i+2i^2}{9-i^2}$
 $= \frac{55-25i}{10}$
 $= \frac{11}{2} - \frac{5i}{2}$

b) $z = -\sqrt{3} + i$

c) $\arg z = \frac{5\pi}{6}$ $|z| = 2$

d) $z^7 + 64z = (2 \operatorname{cis} \frac{5\pi}{6})^7 + 64 \times 2 \operatorname{cis} \frac{5\pi}{6}$
 $= 128 \operatorname{cis} \frac{35\pi}{6} + 128 \operatorname{cis} \frac{5\pi}{6}$
 $= 128 \operatorname{cis} (\frac{\pi}{6}) + 128 \operatorname{cis} \frac{5\pi}{6}$
 $= 128 [\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} + \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}]$
 $= 128 [\frac{\sqrt{3}}{2} - \frac{1}{2}i - \frac{1}{2} + \frac{1}{2}i]$
 $z^7 + 64z = 0$

e) $(2+i)z^2 - 4z + (2-i) = 0$

$z = \frac{4 \pm \sqrt{16 - 4(2+i)(2-i)}}{2(2+i)}$

$= \frac{4 \pm \sqrt{16 - 20}}{2(2+i)}$
 $= \frac{4 \pm \sqrt{-4}}{2(2+i)}$
 $= \frac{4 \pm 2i}{2(2+i)}$
 $= \frac{2 \pm i}{2+i}$
 $\therefore z = \frac{2+i}{2+i} \text{ or } z = \frac{2-i}{2-i}$
 $= 1 \text{ or } z = \frac{3}{5} - \frac{4i}{5}$

Question 8 (begin a new page)

a) Draw neat sketch graphs of the following showing their main features

i) $y = 2 - \sin x$

ii) $y = \ln(2x - 6)$

b) Express $\frac{x^2 - 4x - 1}{(1+x^2)(1+2x)}$ as the sum of two partial fractions.

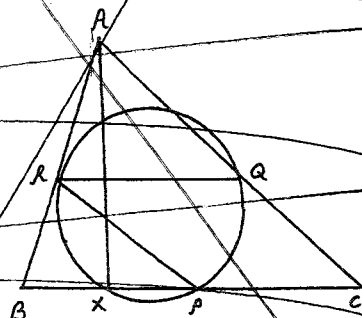
Hence find $\int \frac{x^2 - 4x - 1}{(1+x^2)(1+2x)} dx$

c) P, Q, R are the mid-points of the sides BC, CA and AB of a triangle ABC. The circle through P, Q and R meets the three sides again at X, Y and Z respectively. show that:

i) RPCQ is a parallelogram.

ii) Triangle XQC is isosceles.

iii) AX is perpendicular to BC



END OF PAPER

1999 4 Unit 1/2 Yearly Solutions.

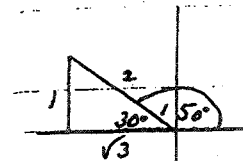
Piii

$$\begin{aligned} Q.1. a) \frac{(2-i)(8+3i)}{3+i} &= \frac{16+6i-8i-3i^2}{3+i} \\ &= \frac{19-2i}{3+i} \times \frac{3-i}{3-i} \\ &= \frac{57-19i-6i+2i^2}{9-i^2} \\ &= \frac{55-25i}{10} \\ &= \frac{11}{2} - \frac{5i}{2} \end{aligned}$$

[4]

b) $z = -\sqrt{3} + i$

i) $\arg z = \frac{5\pi}{6}$ $|z| = 2$



$$\begin{aligned} ii) z^7 + 64z &= \left(2 \operatorname{cis} \frac{5\pi}{6}\right)^7 + 64 \times 2 \operatorname{cis} \frac{5\pi}{6} \\ &= 128 \operatorname{cis} \frac{35\pi}{6} + 128 \operatorname{cis} \frac{5\pi}{6} \\ &= 128 \operatorname{cis} \left(-\frac{\pi}{6}\right) + 128 \operatorname{cis} \frac{5\pi}{6} \\ &= 128 \left[\cos \left(-\frac{\pi}{6}\right) + i \sin \left(-\frac{\pi}{6}\right) + \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right] \\ &= 128 \left[\frac{\sqrt{3}}{2} - \frac{1}{2}i - \frac{\sqrt{3}}{2} + \frac{1}{2}i \right] \\ &= 0 \end{aligned}$$

[5]

$z^7 + 64z = 0$

c) $(2+i)z^2 - 4z + (2-i) = 0$
 $z = \frac{4 \pm \sqrt{16 - 4(2+i)(2-i)}}{2(2+i)}$

[6]

$$\begin{aligned} &= \frac{4 \pm \sqrt{16 - 20}}{2(2+i)} \quad \therefore z = \frac{2+i}{2+i} \text{ or } z = \frac{2-i}{2-i} \times \frac{2-i}{2-i} \\ &= \frac{4 \pm \sqrt{-4}}{2(2+i)} = 1 = \frac{4-4i+i^2}{5} \\ &= \frac{4 \pm 2i}{2(2+i)} = \frac{3-4i}{5} \\ &= \frac{2 \pm i}{2+i} \end{aligned}$$

Sol: $z = 1$ or $z = \frac{3}{5} - \frac{4}{5}i$

$$= (x-1)(x+1)(x-i)(x+i)(x-\sqrt{2})(x+\sqrt{2}) \quad \text{complex}$$

$$p(x) = x^4 + x^2 + 6x + 4$$

$$p'(x) = 4x^3 + 2x + 6$$

$$\text{when } p'(x) = 0 \quad 4x^3 + 2x + 6 = 0$$

$$p'(1) = 4 + 2 + 6 \neq 0$$

$$p'(-1) = -4 - 2 + 6 = 0$$

$$p(-1) = 1 + 1 - 6 + 4 = 0$$

$$p(-1) = 0 \text{ and } p'(-1) = 0 \therefore x = -1 \text{ is double root}$$

$$\therefore (x+1)^2 \text{ is a factor of } p(x)$$

$$x^2 - 2x + 4$$

$$x^2 + 2x + 1 \mid x^4 + x^2 + 6x + 4$$

$$x^4 + 2x^3 + x^2$$

$$-2x^3 + 6x + 4$$

$$-2x^3 - 4x^2 - 2x$$

$$4x^2 + 8x + 4$$

$$4x^2 + 8x + 4$$

$$0$$

$$\therefore p(x) = (x+1)^2 (x^2 - 2x + 4)$$

$$\text{when } x^2 - 2x + 4 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot 4}}{2}$$

$$= \frac{2 \pm \sqrt{-12}}{2}$$

$$= \frac{2 \pm 2\sqrt{3}i}{2} = \frac{2(1 \pm \sqrt{3}i)}{2}$$

$$\therefore \text{zeros of } p(x) \text{ are } x = -1, x = -1, x = 1 \pm \sqrt{3}i$$

$$p(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$$

real coefficients $\therefore$ complex roots are conjugate pairs
roots are $a \pm bi$ and $a \pm 2bi$

$$\text{sum roots } (a+bi) + (a-bi) + (a+2bi) + (a-2bi) = -\frac{b}{a}$$

$$\therefore 4a = 4 \quad a = 1$$

$$\text{product roots } (a+bi)(a-bi)(a+2bi)(a-2bi) = 10$$

$$(a^2 + b^2)(a^2 + 4b^2) = 10 \quad \text{but } a = 1$$

$$\therefore (1 + b^2)(1 + 4b^2) = 10$$

$$1 + 5b^2 + 4b^4 = 10$$

$$4b^4 + 5b^2 - 9 = 0$$

$$(4b^2 + 9)(b^2 - 1) = 0$$

$$4b^2 + 9 = 0$$

$$\text{or } b^2 - 1 = 0$$

no real solⁿ

$$b = \pm 1$$

[6]

$$\therefore a = 1 \quad b = \pm 1$$

$$\therefore \text{zeros of } p(x) \text{ are } (1 \pm i) \text{ and } (1 \pm 2i)$$

$$(x - (1+i))(x - (1-i)) = (x-1-i)(x-1+i)$$

$$= (x-1)^2 + 1$$

$$= (x^2 - 2x + 2)$$

$$(x - (1+2i))(x - (1-2i)) = (x-1-2i)(x-1+2i)$$

$$= (x-1)^2 + 4$$

$$= (x^2 - 2x + 5)$$

$$\therefore p(x) = (x^2 - 2x + 2)(x^2 - 2x + 5)$$

Question 3.

$$a) \quad x^3 + 4x^2 - 3x + 1 = 0$$

$$i) \text{ let new root } x = 2x \quad \text{or } x = \frac{x}{2}$$

$$\text{new eqn } \left(\frac{x}{2}\right)^3 + 4\left(\frac{x}{2}\right)^2 - 3\left(\frac{x}{2}\right) + 1 = 0$$

[3]

$$\frac{x^3}{8} + x^2 - \frac{3x}{2} + 1 = 0$$

$$x^3 + 8x^2 - 12x + 8 = 0$$

$$ii) \text{ let new root } x = \frac{1}{x} \quad \text{or } x = \frac{1}{x}$$

$$\text{new eqn } \left(\frac{1}{x}\right)^3 + 4\left(\frac{1}{x}\right)^2 - 3\left(\frac{1}{x}\right) + 1 = 0$$

$$\frac{1}{x^3} + \frac{4}{x^2} - \frac{3}{x} + 1 = 0$$

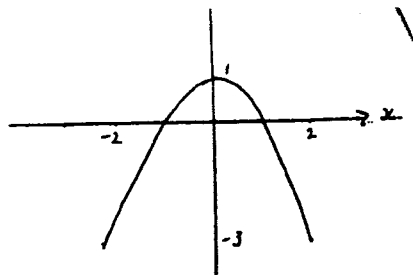
$$1 + 4x - 3x^2 + x^3 = 0$$

[3]

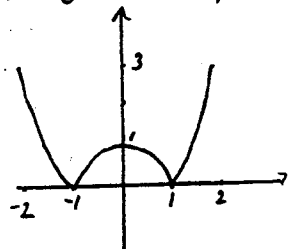
$$x^3 - 3x^2 + 4x + 1 = 0$$

$$f(x) = 1 - x^2 \quad -2 \leq x \leq 2$$

[1]

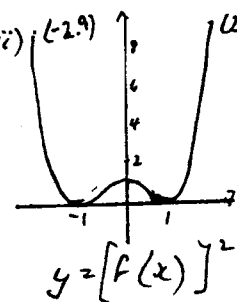


i) $y = |f(x)|$



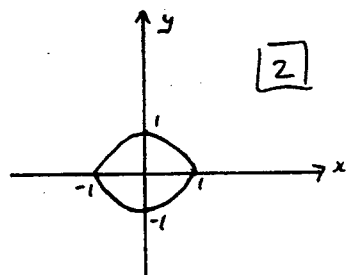
[2]

ii) $y = [f(x)]^2$



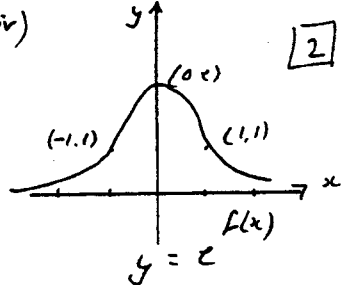
[2]

iii) $|y| = f(x)$



[2]

iv)



[2]

Question 4

1) $1, \omega, \omega^2$ are the roots of $z^3 = 1$
 $\therefore \omega^3 = 1$ and sum of roots of $z^3 - 1 = 0$
 $1 + \omega + \omega^2 = 0$

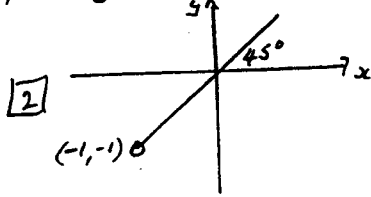
$$\therefore (1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^8) = (1-\omega)(1-\omega^2)(1-\omega)(1-\omega^2) = (1-\omega)^2(1-\omega^2)^2$$

Q4 $(1-\omega)^2(1-\omega^2)^2 = (1-2\omega+\omega^2)(1-2\omega^2+\omega^4)$
 $= (1+\omega^2-2\omega)(1+\omega-2\omega^2)$
 $= (-\omega-2\omega)(-\omega^2-2\omega^2)$
 $= (-3\omega)(-3\omega^2)$
 $= 9\omega^3$
 $= 9$

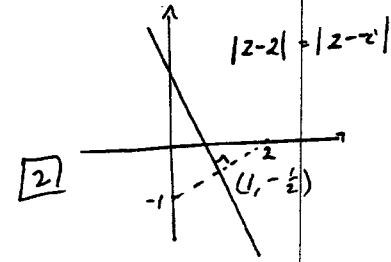
[6]

using $1+\omega+\omega^2=0$

b) $\arg(z - (-1-i)) = \frac{\pi}{4}$



[2]

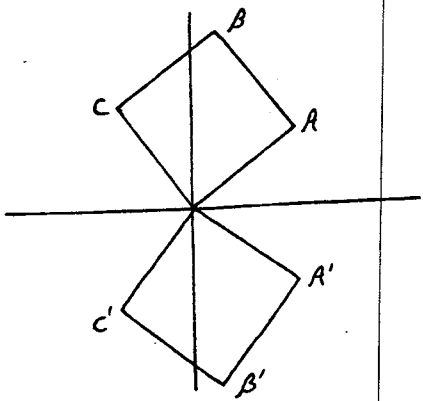


[2]

c) A is z
 C is $i z$
 B is $z + i z$

[6]

A' is $\bar{z}$
 C' is $\overline{i z}$
 B' is $\overline{z + i z}$
 or $\overline{z + i z}$



Question 5

a) $x^2 - 3x + (k-2) = 0$

For distinct real roots $\Delta > 0$

$$(-3)^2 - 4 \times 1 \times (k-2) > 0$$

$$9 - 4k + 8 > 0$$

$$-4k > -17$$

$$k < \frac{17}{4}$$

[2]

(Cont.)
 i) \$50000 at 6% for 20 years
 amount to $50000 (1.06)^{20}$
 $= \$160356.77$

[2]

ii) For 19 years \$1000 is invested annually.
 1st \$1000 will amount to $1000 (1.06)^{19}$
 2nd $1000 (1.06)^{18}$

Final \$1000 $1000 (1.06)^1$

$$\begin{aligned} \text{Total} &= 1000 (1.06)^1 + 1000 (1.06)^2 + \dots + 1000 (1.06)^{19} \\ &= 1000 [1.06 + 1.06^2 + \dots + 1.06^{19}] \\ &= 1000 \cdot \frac{1.06 (1.06^{19} - 1)}{(1.06 - 1)} \\ &= \$35785.59 \end{aligned}$$

$$\begin{aligned} \text{Total money in fund} &= \$160356.77 + \$35785.59 \\ &= \$196142.36 \end{aligned}$$

[4]

i) Area quadrilateral = Square - ΔEBF - ΔAFD

$$\begin{aligned} &= 4 - \frac{1}{2}x^2 - \frac{1}{2} \cdot 2(2-x) \\ &= 4 - \frac{1}{2}x^2 - 2 + x \\ &= 2 + x - \frac{1}{2}x^2 \\ A &= \frac{1}{2}(4 + 2x - x^2) \end{aligned}$$

[4]

ii) $\frac{dA}{dx} = \frac{1}{2}(2 - 2x)$
 when $\frac{dA}{dx} = 0$ $2 - 2x = 0$
 $x = 1$

Maximum turning point
 at $x = 1$

$$A = \frac{1}{2}(4 + 2 - 1)$$

Area of quadrilateral is $2\frac{1}{2}$ sq units

x	1	1	1
$\frac{dA}{dx}$	+	0	-
		Max	

Q6. Vol. rect. slice = $2y \times 4y \cdot \Delta x$
 $= 8y^2 \cdot \Delta x$

$$\therefore \text{Vol solid} = \sum_{\Delta x \rightarrow 0} 8y^2 \Delta x$$

$$= 2 \int_0^4 8y^2 dx$$

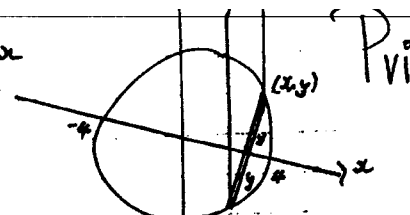
$$= 2 \cdot 8 \int_0^4 (16 - x^2) dx$$

$$= 16 \left[16x - \frac{1}{3}x^3 \right]_0^4$$

$$= 16 \left[16 \times 4 - \frac{1}{3}4^3 \right]$$

$$= 16 \times \frac{2}{3} \times 64$$

$$\text{Vol solid} = \frac{2048}{3} \text{ cu units}$$



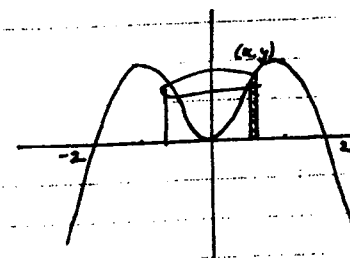
b)

$$y = 4x^2 - x^4$$

$$4x^2 - x^4 = 0$$

$$x^2(4 - x^2) = 0$$

$$x^2(2-x)(2+x) = 0$$



Vol shell = $2\pi r h \times \text{thickness}$

$$= 2\pi xy \Delta x$$

$$\text{Vol solid} = \sum_{\Delta x \rightarrow 0} 2\pi xy \Delta x$$

$$= 2\pi \int_0^2 x(4x^2 - x^4) dx$$

$$= 2\pi \int_0^2 (4x^3 - x^5) dx$$

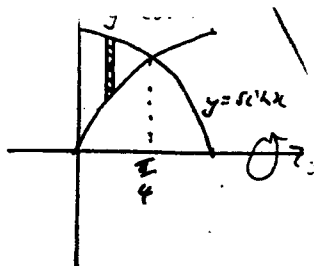
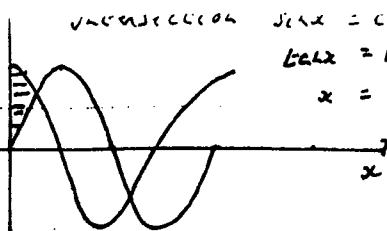
$$= 2\pi \left[x^4 - \frac{1}{6}x^6 \right]_0^2$$

$$= 2\pi \left[16 - \frac{1}{6} \times 64 - 0 \right]$$

$$\text{Vol} = \frac{32\pi}{3} \text{ cu units}$$

1. C.

Intersection $\sin x = \cos x$
 $\tan x = 1$
 $x = \frac{\pi}{4}$



$$\text{Vol disc} = (\pi \cos^2 x - \pi \sin^2 x) \Delta x$$

$$\text{Vol solid} = \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) \Delta x$$

$$= \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) dx$$

$$= \pi \int_0^{\pi/4} \cos 2x \cdot dx$$

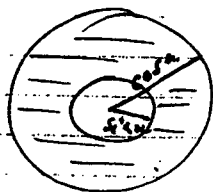
$$= \pi \left[\frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$= \pi \left[\frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0 \right]$$

$$= \frac{\pi}{2} \sin \frac{\pi}{2}$$

$$= \frac{\pi}{2} \times 1$$

$$\text{Volume} = \frac{\pi}{2} \text{ cu. units.}$$



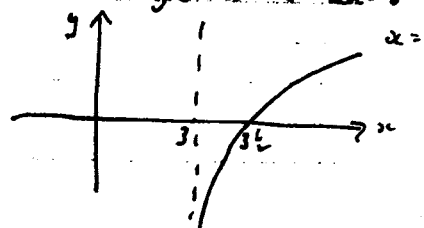
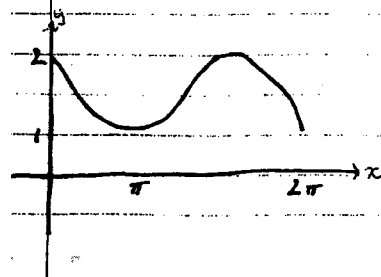
7 See 2 unit solutions.

P $y = 2 - \sin x$

$y = 6(2x - 6)$

asymptote when $2x - 6 = 0$
 $x = 3$

x intercept when $2x - 6 = 0$
 $x = 3$



8b) $\frac{x^2 - 4x - 1}{(1+x^2)(1+2x)} = \frac{Ax+B}{1+x^2} + \frac{C}{1+2x}$ Pvii
 $\therefore x^2 - 4x - 1 = (Ax+B)(1+2x) + C(1+x)$

$x = -\frac{1}{2} \quad \frac{1}{4} + 2 - 1 = 0 + C(1 + \frac{1}{2})$
 $\frac{1}{4} = \frac{3}{4}C \quad C = 1$

$x = 0 \quad -1 = B + C \quad \text{but } C = 1 \therefore B = -2$

$x = 1 \quad 1 - 4 - 1 = (A-2)(1+2) + 1(1+1)$

$-4 = 3A - 6 + 2$

$0 = 3A \quad \therefore A = 0$

$\therefore \frac{x^2 - 4x - 1}{(1+x^2)(1+2x)} = \frac{-2}{1+x^2} + \frac{1}{1+2x}$

$\int \frac{x^2 - 4x - 1}{(1+x^2)(1+2x)} dx = \int \frac{-2}{1+x^2} + \frac{1}{1+2x} dx$
 $= -2 \tan^{-1} x + \frac{1}{2} \ln(1+2x) + C$

c) In ΔARQ & ΔBAC

angle A is common

$\frac{AR}{AB} = \frac{AQ}{AC} = \frac{1}{2}$ (R & Q are midpts AB & AC)

$\therefore \Delta ARQ \sim \Delta BAC$ [2 pairs sides in same ratio and same included angle]

$\therefore \angle AQR = \angle ACB$ (corresponding angles Δ s)

$\therefore RQ \parallel BC$ (corresponding angles equal)

Similarly $RP \parallel AC$

$\therefore RPQ$ is a parallelogram (2 pairs opp sides)

$\angle QRP = \angle QXP$ (angle in same segment or $\parallel$ (parallels))

$\angle QRP = \angle QCP$ (opp angles $\parallel$)

$\therefore \angle QXP = \angle QCP$

$\therefore \Delta XQC$ is isosceles.

$$\lim_{n \rightarrow \infty} \frac{\log n}{n} = 0$$

$$\lim_{n \rightarrow -\infty} ne^n = 0$$

(3)

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\tan^{-1}(n+1) - \tan^{-1}n = \alpha - \beta \text{ say}$$

$$\begin{aligned} \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{n+1 - n}{1 + n(n+1)} \\ &= \frac{1}{1 + n + n^2} \end{aligned}$$

$$\therefore \cot(\alpha - \beta) = n^2 + n + 1 \quad (2)$$

$$\tan^{-1}(n+1) - \tan^{-1}n = \cot^{-1}(n^2 + n + 1)$$

$$\text{find } \cot^{-1}1 + \cot^{-1}3 + \dots + \cot^{-1}31$$

$$\text{let } n = 0,$$

$$\tan^{-1}1 - \tan^{-1}0 = \cot^{-1}1$$

$$= 1, \tan^{-1}2 - \tan^{-1}1 = \cot^{-1}3$$

$$= 5, \tan^{-1}6 - \tan^{-1}5 = \cot^{-1}31$$

$$\cot^{-1}1 + \cot^{-1}3 + \dots + \cot^{-1}31$$

$$(\tan^{-1}1 - \tan^{-1}0) + (\tan^{-1}2 - \tan^{-1}1) + \dots + (\tan^{-1}6 - \tan^{-1}5)$$

$$\tan^{-1}6 - \tan^{-1}0$$

$$= \tan^{-1}6$$

(1)

$$\begin{aligned} (1) f(x) &= \sin x \\ f'(x) &= \cos x \\ f''(x) &= -\sin x \\ f'''(x) &= -\cos x \\ f^{(4)}(x) &= \sin x \end{aligned}$$

$$\begin{aligned} f(0) &= 0 \\ f'(0) &= 1 \\ f''(0) &= 0 \\ f'''(0) &= -1 \\ f^{(4)}(0) &= 0 \end{aligned}$$

$$\therefore \sin x = 1x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$(1) f(x) = \cos x$$

$$f(0) = 1$$

$$\begin{aligned} f'(x) &= -\sin x \\ f''(x) &= -\cos x \\ f'''(x) &= \sin x \\ f^{(4)}(x) &= +\cos x \end{aligned}$$

$$\begin{aligned} f'(0) &= 0 \\ f''(0) &= -1 \\ f'''(0) &= 0 \\ f^{(4)}(0) &= 1 \end{aligned}$$

$$\therefore \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$(1) f(x) = e^{ix}$$

$$f(0) = 1$$

$$\begin{aligned} f'(x) &= ie^{ix} \\ f''(x) &= -e^{ix} \\ f'''(x) &= -ie^{ix} \\ f^{(4)}(x) &= e^{ix} \end{aligned}$$

$$\begin{aligned} f'(0) &= i \\ f''(0) &= -1 \\ f'''(0) &= -i \\ f^{(4)}(0) &= 1 \end{aligned}$$

$$\therefore e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \dots$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$$

$$e^{ix} = \cos x + i \sin x$$

$$e^{i\pi} = \cos \pi + i \sin \pi$$